

A Method of Multiple Targets DOA Tracking with Vector Hydrophone Based on Unite Processing in Time and Frequency Domain

Wei Zhang
Divison 105

Yichang Research Institute of
Testing Technology
Yichang, China
zhangwei667@163.com

Ling Shang
Metrology Department

Yichang Research Institute of
Testing Technology
Yichang, China
64505513@qq.com

Hongxing Tian
Divison 105

Yichang Research Institute of
Testing Technology
Yichang, China
1574503594@qq.com

Yangbo Zhang
Divison 105

Yichang Research Institute of
Testing Technology
Yichang, China
zhangyangbo818@163.com

Abstract—Acoustic pressure and particle velocity is narrowband filtered to obtain multi-band signal. High-order correlation equations composed of acoustic pressure and particle velocity are solved by optimization algorithm and DOA (Direction of arrive) distribution for different signal in multi-band is counted. Then the DOA of multiple targets can be distinguished and tracked. In order to improve the accuracy, the least square method is used to fit the measurement results and a prediction model is established, and then DOA trajectory is optimized by Kalman filter. The experimental results indicate that DOA of three targets can be distinguished and trajectory tracked, DOA of interference source is also observed by the pseudo color picture of DOA distribution, the accuracy of DOA estimation is less than 5° .

Keywords—vector hydrophone, DOA estimation, least square method, Kalman filter, optimization algorithm

I. INTRODUCTION

The detection of target radiated noise and passive DOA estimation are always an important question for discussion of under water early warning[1]. With the development of vibration and noise reduction, radiated noise of underwater target reduced year by year, especially in the high frequency band. In order to detect long-distance targets, it is generally necessary to carry out in the low frequency band. For low frequency signals, in order to obtain ideal spatial gain, the traditional sound pressure array must have a large physical aperture, which is inconvenient or even impossible in engineering application. Vector hydrophone can be output acoustic pressure and particle velocity at the same point synchronously. Single vector hydrophone can detect DOA without ambiguity in the whole space[2], which makes it has unique advantages in underwater acoustic detection, especially on small-scale platforms[3].

The traditional application methods of DOA estimation using single vector hydrophone in engineering include average sound intensity meter[4][5], cross spectrum method[6][7] and so on. The method of average sound intensity meter is based on maximum likelihood estimation of sound energy flow, which

can suppress the isotropic interference effectively. However, if there are multiple targets, it measures the vector sum of the sound energy flow of all signals, so it cannot distinguish the different targets. Cross spectrum method is a frequency domain processing method, which has the ability of distinguishing multiple targets. However, it cannot do nothing about targets whose characteristic spectra overlap or interfere with each other. The literature[8][9] shows that for incoherent targets, when the SNR is enough and the line spectrum characteristics are uncertain, the number of distinguished targets can reach 7 by making full use of the correlation characteristics of acoustic pressure and particle velocity in theory. However, The experimental results in literature[10] indicate that this method is not effective when two targets are close.

In this paper, a method of multiple targets DOA tracking by using single Vector hydrophone based on unite processing in time and frequency domain is proposed. Firstly, the relationship between acoustic pressure and particle velocity is solved in each frequency band, and the high-order correlation equations are solved to obtain the DOA of multiple targets. Then the least square method is used to fit the DOA trajectory in a period of time and establish a dynamic prediction model. Finally, Kalman filter is performed and the DOA distribution of targets in each frequency band is counted to obtain the pseudo color picture of DOA distribution. By using this method, we can not only distinguish the DOA trajectory of multiple targets, but also observe the azimuth of interference. The accuracy of DOA estimation is less than 5° .

II. HIGH-ORDER CORRELATION EQUATIONS

The acoustic signal sent by each independent target and reaching the receiving point is S_i , the horizontal azimuth angle is θ_i , subscript is used to distinguish different targets. In far field conditions, $kr \gg 1$, k is wave number, r is horizontal distance. Acoustic pressure p and two dimensional horizontal particle velocity v_x , v_y are as follows

$$\begin{cases} p = \sum_{i=1}^N s_i(t) + n_p(t) \\ v_x = \sum_{i=1}^N s_i(t) \cos \theta_i / (\rho c) + n_x(t) \\ v_y = \sum_{i=1}^N s_i(t) \sin \theta_i / (\rho c) + n_y(t) \end{cases} \quad (1)$$

where, N is the number of targets, n_p , n_x , n_y are the corresponding background noise respectively, ρ is density of medium, c is sound speed.

In the condition of isotropic noise, the correlation equations between acoustic pressure and particle velocity can be established. If SNR is enough, because of independence of noise and signal, the correlation between signal and noise and between noises can be ignored. Thus, the following equations can be obtained

$$\langle p(t), v_x(t) \rangle = \sum_{i=1}^N I_i \cos \theta_i / (\rho c) \quad (2)$$

$$\langle p(t), v_y(t) \rangle = \sum_{i=1}^N I_i \sin \theta_i / (\rho c) \quad (3)$$

$$\langle v_x(t), v_y(t) \rangle = \sum_{i=1}^N I_i \sin \theta_i \cos \theta_i / (\rho c)^2 \quad (4)$$

$$\langle v_x(t), v_x(t) \rangle = \sum_{i=1}^N I_i \cos^2 \theta_i / (\rho c)^2 \quad (5)$$

$$\langle v_y(t), v_y(t) \rangle = \sum_{i=1}^N I_i \sin^2 \theta_i / (\rho c)^2 \quad (6)$$

$$2\langle p(t), p(t) \rangle^2 - \langle p^2(t), p^2(t) \rangle = \sum_{i=1}^N I_i^2 \quad (7)$$

$$\begin{aligned} & 2\langle p(t), p(t) \rangle \langle p(t), v_x(t) \rangle - \langle p^2(t), p(t) v_x(t) \rangle \\ &= \sum_{i=1}^N I_i^2 \cos \theta_i / (\rho c) \end{aligned} \quad (8)$$

$$\begin{aligned} & 2\langle p(t), p(t) \rangle \langle p(t), v_y(t) \rangle - \langle p^2(t), p(t) v_y(t) \rangle \\ &= \sum_{i=1}^N I_i^2 \sin \theta_i / (\rho c) \end{aligned} \quad (9)$$

$$\begin{aligned} & 2\langle p(t), v_x(t) \rangle \langle p(t), v_y(t) \rangle - \langle p^2(t), v_x(t) v_y(t) \rangle \\ &= \sum_{i=1}^N I_i^2 \sin \theta_i \cos \theta_i / (\rho c)^2 \end{aligned} \quad (10)$$

$$\begin{aligned} & 2\langle p(t), v_y(t) \rangle^2 - \langle p^2(t), v_y^2(t) \rangle \\ &= \sum_{i=1}^N I_i^2 \sin^2 \theta_i / (\rho c)^2 \end{aligned} \quad (11)$$

$$\begin{aligned} & 2\langle p(t), v_x(t) \rangle \langle v_x(t), v_y(t) \rangle - \langle p(t) v_y(t), v_x^2(t) \rangle \\ &= \sum_{i=1}^N I_i^2 \sin \theta_i \cos^2 \theta_i / (\rho c)^3 \end{aligned} \quad (12)$$

$$\begin{aligned} & 2\langle p(t), v_y(t) \rangle \langle v_y(t), v_x(t) \rangle - \langle p(t) v_x(t), v_y^2(t) \rangle \\ &= \sum_{i=1}^N I_i^2 \sin^2 \theta_i \cos \theta_i / (\rho c)^3 \end{aligned} \quad (13)$$

$$\begin{aligned} & 2\langle v_x(t), v_x(t) \rangle \langle v_x(t), v_y(t) \rangle - \langle v_x^2(t), v_x(t) v_y(t) \rangle \\ &= \sum_{i=1}^N I_i^2 \sin \theta_i \cos^3 \theta_i / (\rho c)^4 \end{aligned} \quad (14)$$

$$\begin{aligned} & 2\langle v_x(t), v_y(t) \rangle^2 - \langle v_x^2(t), v_y^2(t) \rangle \\ &= \sum_{i=1}^N I_i^2 \sin^2 \theta_i \cos^2 \theta_i / (\rho c)^4 \end{aligned} \quad (15)$$

$I_i = \langle s_i(t), s_i(t) \rangle$ is the sound source intensity of each target.

Although according to the arrangement and combination, other forms of equations can also be written. However, it can be proved that the second-order moment and the forth-order moment can only have 14 independent equations. Because there are two unknown parameter for each target, I and θ , if the number of targets is less than 7, the sound source intensity and horizontal azimuth angle of all targets can be counted by solving equations. Because those equations are nonlinear and the results can not be expressed as precise mathematical expression, they must be solved by using optimization algorithm.

III. KALMAN FILTER

The system with discrete control process can be described by linear differential equation

$$\mathbf{X}(k) = \mathbf{A} \mathbf{X}(k-1) + \mathbf{B} \mathbf{U}(k) + \mathbf{W}(k) \quad (16)$$

The measurement result is

$$\mathbf{Z}(k) = \mathbf{H} \mathbf{X}(k) + \mathbf{V}(k) \quad (17)$$

Where, $\mathbf{X}(k)$ and $\mathbf{U}(k)$ are system state vector and control vector at moment k respectively. $\mathbf{A}$ and $\mathbf{B}$ are the parameters of system. $\mathbf{Z}(k)$ is the measurement result, $\mathbf{H}$ is the parameter of measurement system. $\mathbf{W}(k)$ and $\mathbf{V}(k)$ are the Gauss white noise of process and measurement respectively, $\mathbf{Q}$ and $\mathbf{R}$ are the corresponding variance.

Assume that the current system state is $\mathbf{k}$, then it can be predicted according to the last moment.

$$\mathbf{X}(k|k-1) = \mathbf{A} \mathbf{X}(k-1|k-1) + \mathbf{B} \mathbf{U}(k) \quad (18)$$

Where, $\mathbf{X}(k|k-1)$ is predicted current result according to the last moment, $\mathbf{X}(k-1|k-1)$ is the optimal result at the last moment.

$\mathbf{P}(k-1|k-1)$ and $\mathbf{P}(k|k-1)$ are mean square error of $\mathbf{X}(k-1|k-1)$ and $\mathbf{X}(k|k-1)$ respectively. They can be expressed as

$$\mathbf{P}(k|k-1) = \mathbf{A} \mathbf{P}(k-1|k-1) \mathbf{A}' + \mathbf{Q} \quad (19)$$

Combined with the measurement result, the optimal estimation value at the the current moment can be obtained after counting predicted state of system.

$$\mathbf{X}(k|k) = \mathbf{X}(k|k-1) + \mathbf{K}_g(k) (\mathbf{Z}(k) - \mathbf{H} \mathbf{X}(k|k-1)) \quad (20)$$

Where, $\mathbf{K}_g$ is the gain from Kalman filter and it is expressed as

$$\mathbf{K}_g(k) = \mathbf{P}(k|k-1) \mathbf{H}' / (\mathbf{H} \mathbf{P}(k|k-1) \mathbf{H}' + \mathbf{R}) \quad (21)$$

In order to the filter can run until the end, it is necessary to update $\mathbf{P}(k|k)$

$$\mathbf{P}(k|k) = (\mathbf{I} - \mathbf{K}_g(k) \mathbf{H}) \mathbf{P}(k|k-1) \quad (22)$$

Where, $\mathbf{I}$ is identity matrix.

Formula(18)~ (22) are the description of the Kalman filter process.

IV. SOLUTION FLOW OF MULTIPLE TARGETS DOA TRACKING

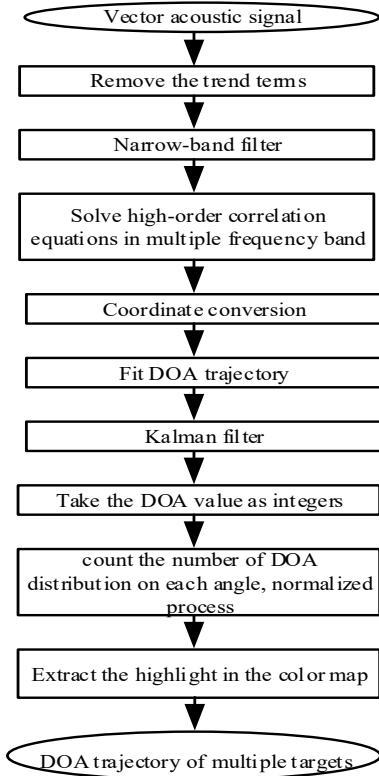


Fig. 1. Solution flow of multiple targets DOA tracking

As shown in Figure 1, firstly, the trend terms of signal are removed to eliminate DC and low- frequency interference, and then narrow-band filter is carried out. The DOA of each target relative to the hydrophone coordinate is obtained by solving

high-order correlation equations in multiple frequency band. The DOA relative to the geodetic coordinate is counted by coordinate conversion. The least square method is used to fit DOA trajectory of targets, and then the outliers are eliminated by Kalman filter. Take the DOA value as integers, count the number of DOA distribution on each angle, normalized process, and draw pseudo color picture according to the distribution. Extract the highlight in the pseudo color picture and fit the DOA trajectory, so as to realize the DOA trajectory of multiple targets tracking.

V. EXPERIMENTAL DATA PROCESSING

The targets are three sightseeing boats. The location relationship and the distance between targets and vector hydrophone are shown in Fig 2.

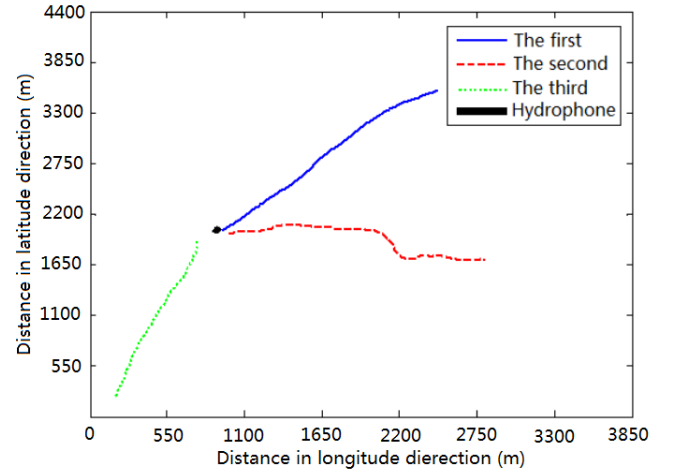


Fig. 2. The location relationship between targets and vector hydrophone

The pseudo color picture of DOA trajectory estimation of three targets is shown in Fig 3.

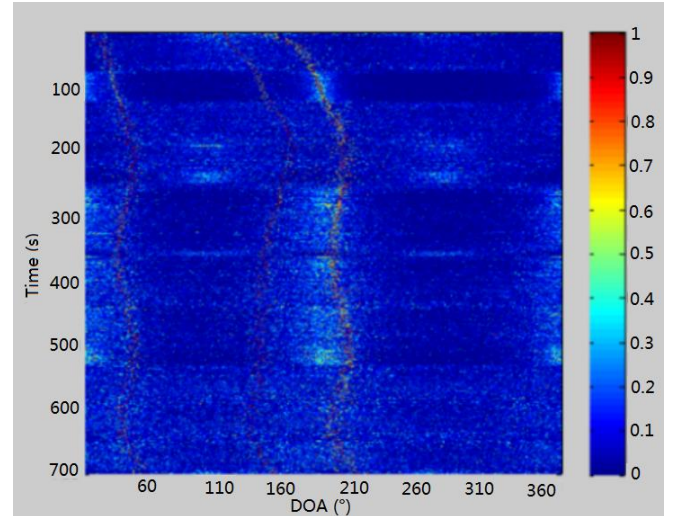


Fig. 3. The pseudo color picture of DOA trajectory

After highlight extraction, the DOA of three targets is obtained, as shown in Fig 4~Fig 6.

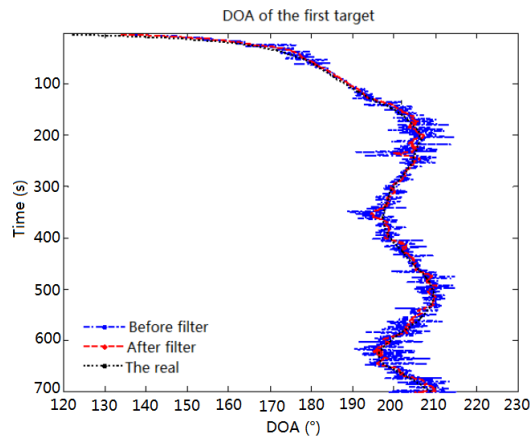


Fig. 4. DOA estimation of the first target

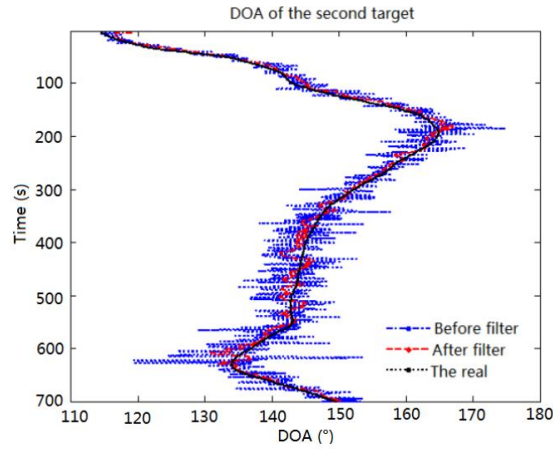


Fig. 5. DOA estimation of the second target

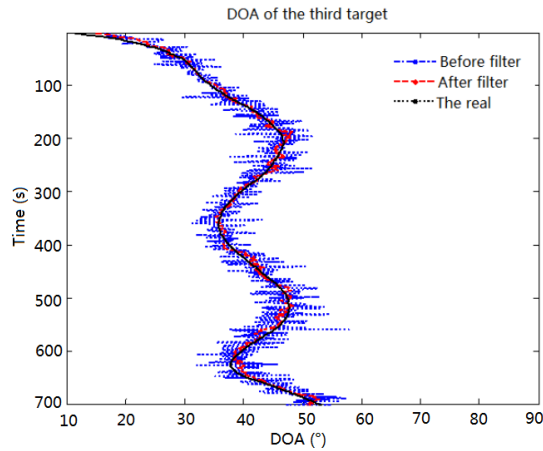


Fig. 6. DOA estimation of the third target

The DOA estimation errors before and after Kalman filter are shown in Fig 7~Fig 8 respectively. It can be seen that before kalman filtering, the maxim error is 18° , and after kalman filtering, the error is less than 5° . It should be noted that the error after kalman filtering at begin point is larger, the reason is that those data are not involved in filtering as reference points. In

addition, the error tends to increase with increase of distance, which is due to the gradual reduction of SNR.

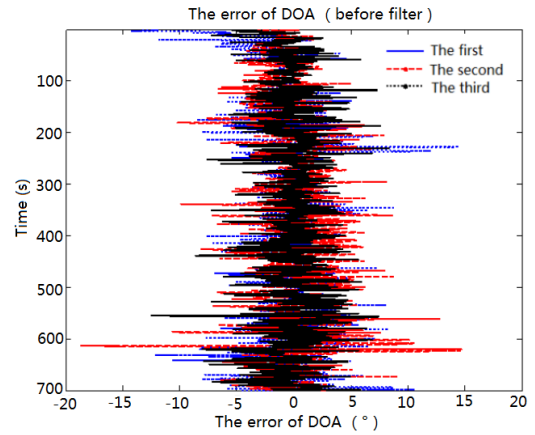


Fig. 7. The error of DOA before filter

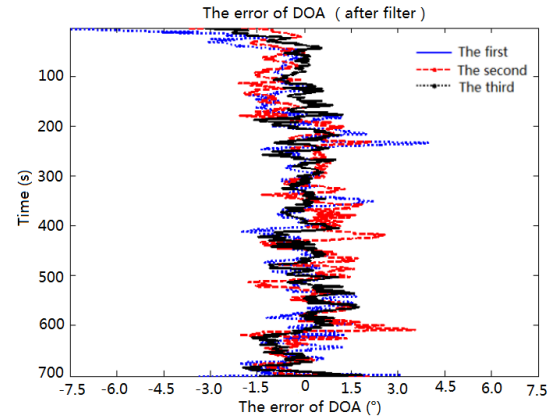


Fig. 8. The error of DOA after filter

VI. CONCLUSION

A new method of multiple targets DOA tracking with vector hydrophone is proposed in this paper. The experiment results indicate that the DOA information of multiple targets can be solved and tracked stably. The azimuth of interference can also be obtained though DOA pseudo color picture. In addition, the outliers can be eliminated and the accuracy can be improved by the method of fitting DOA data and Kalman filter. In the experiment, the accuracy can reach 5° .

REFERENCES

- [1] Ma Lin, Zhao Anbang, Wang Keren, "Vector hydrophone combined channel post adaptive processing and its application". Journal of Harbin Engineering University, 2020, 41(7), pp.991-997.
- [2] Ma Bole, Zhu Shiqiang, Sun Guiqing, "Single vector hydrophone target detection based on eigenvalue". Journal of national university of defense technology, 2019, 41(1), pp.95-100.
- [3] Wang Wenlong, Wang Chao, HAN Mei. "Research on Application of Vector Hydrophone onboard an Underwater Glider". ACTA ARMAMENTARII, 2019, 40(11), pp.2580-2586.
- [4] Yao Zhixiang, Hui Junying, Ying Jingwei, "Four approach to DOA estimation based on signal vector hydrophone". The Ocean Engineering, 2006, 24(1), pp.122 -127.

- [5] Liu Bosheng, Tian Baojing, "Simulation study on errors for evaluation of target azimuth by single vector transducer". Journal of Harbin Engineering University, 2003, 24(5), pp.491-495.
- [6] Zhang Liang, Tian Tian, Meng Chunxia, "Identifying multiple targets by a single vector hydrophone based on the cross-spectrum goniometry". Ship Science and Technology, 2009,31(10), pp.18- 20.
- [7] Sun Xiangqian, Fan Zhan, Li Qing, "A method of target DOA estimation based on correlation integral and cross spectrum WVD". Technical Acoustics 2015, 34(1), pp.23-28
- [8] Cheng Bingbing, "Research on DOA estimation based on two-dimension with an pressure-gradient vector hydrophone". Xi'an, Northwestern Polytechnic University, 2007.
- [9] Yang Shi'e, "Method of multi-sources distinguishing by single Vector transducer". Journal of Harbin Engineering University, 2003, 24(6) pp.591-595.
- [10] Zhang Wei, Meixinhua, Li Caiju, "DOA estimation of multiple sources underwater with single vector hydrophone", MINE WARFARE & SHIP SELF-DEFENCE 2017,25(3), pp.1-6.